Project Title

Intelligent Multilingual Spell Checking Suggestion System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Intelligent Spell Checker Using Levenshtein Edit Distance** is designed to detect spelling errors and provide accurate correction suggestions using string matching and dynamic programming techniques. The system compares user-input words with words stored in a dictionary and calculates the minimum number of insertions, deletions, and substitutions required to transform an incorrect word into a valid word. Based on the calculated edit distances, the system ranks possible corrections and displays the most relevant suggestions to the user.

The application uses dynamic programming to improve computational efficiency by storing intermediate results and avoiding repeated calculations. It can be optimized to handle large dictionaries through efficient searching, candidate filtering, and reduced unnecessary comparisons. The system also supports Unicode text and can be extended to multiple languages by using appropriate language-specific dictionaries and text-processing methods. Additional improvements, such as word-frequency-based ranking and context-aware suggestions, can further enhance correction accuracy.

The Intelligent Spell Checker can be applied in word processors, educational platforms, search engines, email services, messaging applications, and mobile applications. It helps users minimize spelling mistakes, improve writing quality, and communicate more effectively. This project demonstrates the practical application of dynamic programming, string matching, algorithm optimization, and multilingual text processing in developing an efficient and user-friendly spelling correction system.

Signature of the Guide